

# A1: Computer Fundamentals - Exam Lab



This exam practice page is designed for the A1: Computer Fundamentals unit of the new IB Computer Science syllabus (first assessment 2027).

## Syllabus Links:

**A 1.1 Computer Hardware and Operation**

**A 1.2 Data representation and computer logic**

**A 1.3 Operating systems and control systems**

**A 1.4 Translation (HL only)**

"If a computer can only understand 'on' or 'off' (1 or 0), how does it manage to run a complex weather simulation and a web browser at the same time?"

## The Exam Scenario

**Scenario:** The Research Simulation Hub

A university research lab uses a high-performance cluster to run complex weather simulations. The system utilises multiple multi-core CPUs and advanced GPUs, coordinated by a custom-built operating system.

## Core Questions (HL and SL)

Give these questions a try! If you're stuck, check out the 'Strategy' section to help you get started. Once you're finished, use the Mark Scheme and Worked Example to review your progress. If these feel a bit tough right now, head back to the eBook pages to sharpen your understanding first.

### Question 1: Hardware Interaction (5 Marks)

(a) **Identify** the two main units of the central processing unit (CPU). **[2 marks]**

(b) **Outline** the role of the Program Counter (PC) register during the fetch-decode-execute cycle. **[3 marks]**

**Level 1: Strategy**

**Level 2: Mark Scheme**

**Level 3: Worked Example**

### Question 2: Data & Logic (5 Marks)

A safety-critical sensor in the lab uses the following logic circuit to trigger an alarm:

$$Z = (A \text{ AND } B) \text{ OR } (\text{NOT } C)$$

(a) **State** the binary equivalent of the hexadecimal number **AC**. **[1 mark]**

(b) **Construct** a truth table for the Boolean expression:  $Z = (A \text{ AND } B) \text{ OR } (\text{NOT } C)$ . **[4 marks]**

**Level 1: Strategy**

**Level 2: Mark Scheme**

**Level 3: Worked Example**

### Question 3: Operating Systems (4 Marks)

The lab's operating system manages many simultaneous sensor inputs. **Compare and contrast** the use of **polling** and **interrupt handling** for managing data from these sensors. **[4 marks]**

**Level 1: Strategy**

## Level 2: Mark Scheme

## Level 3: Worked Example

# HL Only: Advanced Architecture & Translation

Try to answer the questions if you need help, and have a look at the Strategy to help you get started. Once you have answered the question, you can then check the Mark Scheme and the Worked Example. If you find the questions too hard right now, then go back to the eBook pages first to develop your understanding.

## Question 4: Advanced Hardware (5 Marks)

(a) The hub's multi-core CPUs use **pipelining**. **Describe** how the **write-back stage** of pipelining improves system performance. **[3 marks]**

(b) **Explain** the design philosophy differences between a CPU and a GPU regarding **memory access** and **power efficiency**. **[2 marks]**

## Level 1: Strategy

## Level 2: Mark Scheme

## Level 3: Worked Example

## Question 5: Resource Management & Multitasking (6 Marks)

(a) During a simulation, the OS must manage **resource contention**. **Explain** how a **deadlock** occurs in this environment and **suggest** one way the OS might prevent it. **[4 marks]**

(b) **Distinguish** between an **open-loop** and a **closed-loop** control system within the lab's cooling infrastructure. **[2 marks]**

## Level 1: Strategy

## Level 2: Mark Scheme

## Level 3: Worked Example

## Question 6: The Translation Process (6 Marks)

The lab developers are choosing between **Just-In-Time (JIT) compilation** and a traditional **Interpreter**.

(a) **Evaluate** the use of a **bytecode interpreter** for a simulation that needs to be **cross-platform**.  
[6 marks]

**Level 1: Strategy**

**Level 2: Mark Scheme**

**Level 3: Worked Example**

## How confident do you feel with this topic?

**Exam Ready:** I answered most questions correctly without looking at the eBook. I can explain these concepts to someone else and understand the "why" behind the mark scheme.

*Move on to the next topic.*

**Practitioner:** I understood the general concepts, but I missed a few marks on specific terminology or complex applications. I needed the "Strategy" guide for the harder questions.

*Annotate your answers with the Mark Scheme and revisit the eBook sections where you lost marks. You can also use the bookmarking and notes feature on the eBook pages to highlight certain areas.*

**Novice:** I struggled to get started on the questions. I found it difficult to apply the theory to the practice scenarios.

*Go back to the eBook and Worked Examples before attempting these questions again. Perhaps use some of the Quizlets to practice the key terms and review the command terms.*

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